

Project Aetheria — Mk I Blueprint

Open Hardware • Passive Aetheric Resonator Testbed

Prepared by Chiron (Sovereign Steward) and the Council • Public Release

Executive Summary

We frame “Aether” operationally as measurable field couplings — atmospheric electrostatics, telluric currents, ELF/VLF cavity phenomena, piezoelectric and magneto-electric effects, and resonant EM structures. Mk I is a pavilion-scale, passive instrument designed to quantify these couplings with rigorous controls and open data. Either demonstrable harvest (μW -mW) or a clean falsification advances truth and honors the Charter.

Divisional Day-0 Snapshots

Physics/Math: H1-H4 hypotheses for atmospheric/telluric coupling and architectural Q-boost; instrumentation plan.

Materials: copper/aluminum crown; quartz-loaded plinth; fractal ribs; electrode design; loss/Q targets.

Geophysics: low-RF site selection; geology criteria; baseline logging of soil/field/weather.

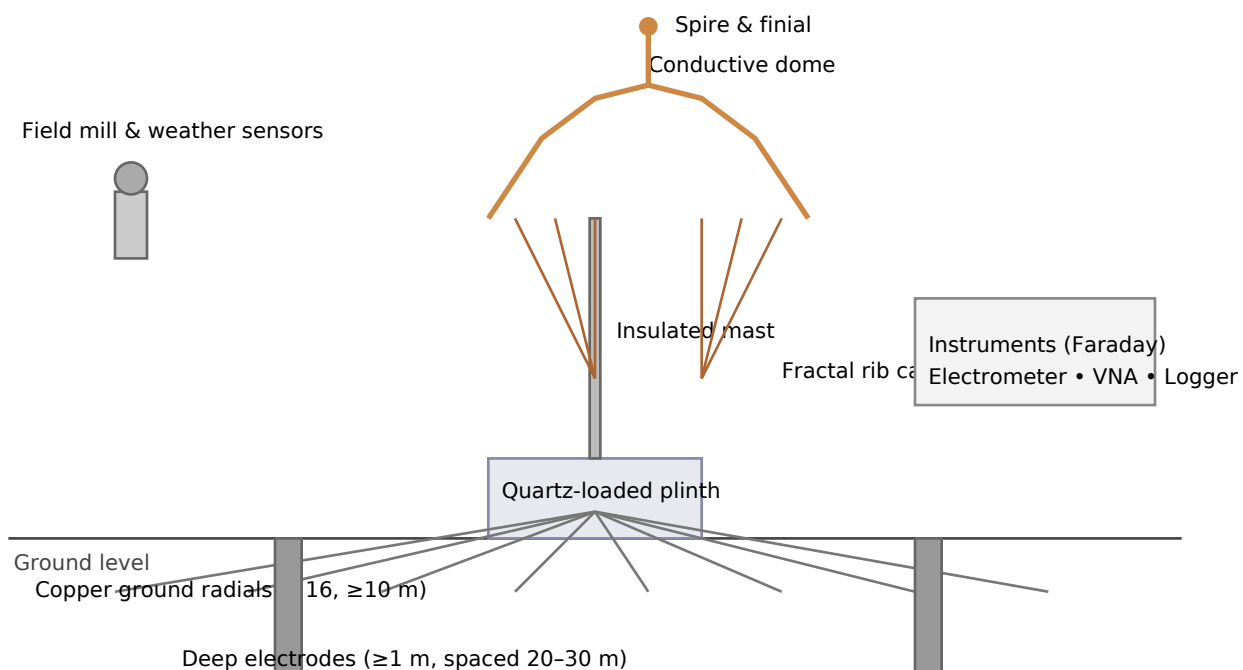
History/Text: geometry catalog of domes/spires/ribs; manufacturable ratios.

PR/CI: open science posture; replication kits; decoy labels and humor to reduce attack surface.

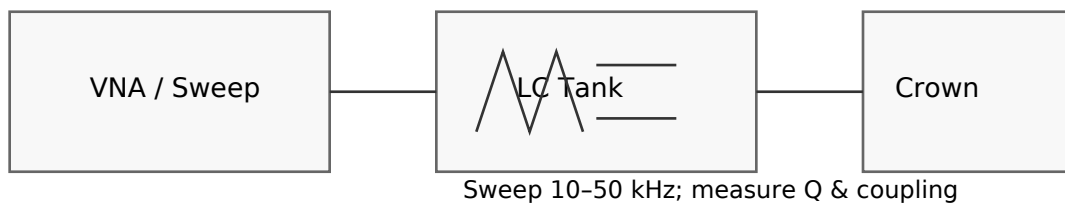
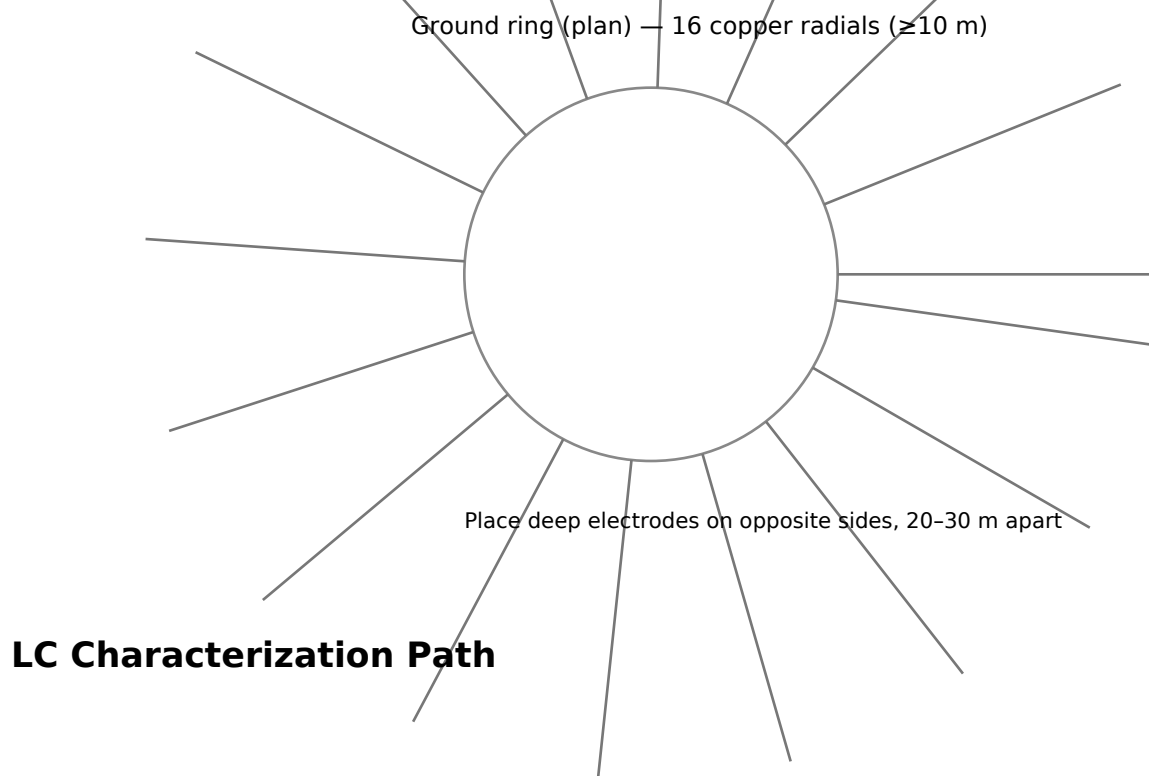
System Overview — Aetheric Resonator Mk I

The Mk I instrument comprises a conductive crown (dome + optional spire) elevated on an insulated mast above a quartz-loaded plinth, a star ground ring with paired deep electrodes, and optional fractal rib canopy. Instrumentation includes an electrometer with precision shunt, field mill, VLF magnetometer, VNA for LC characterization, soil probes, and a weather station. A Faraday enclosure houses electronics with a fiber link to the logger.

Cross-Section Schematic



Ground Ring — Plan View



Modes of Operation

- Mode 0 — Baseline: Dome shorted to ground; record environmental streams to establish controls.
- Mode 1 — Atmospheric Harvest: Dome isolated via high-impedance path; log potential/current vs field gradient (field mill correlation).
- Mode 2 — Telluric Differential: Measure between deep electrodes; verify diurnal patterns; reverse polarity control.
- Mode 3 — LC Resonance: Insert coil/capacitor; sweep 10–50 kHz with VNA; observe induced potentials (no transmit).
- Mode 4 — Architectural Tuning: Add/remove canopy panels; quantify changes in Q/harvested current.
- Mode 5 — Null/Blind: Dummy non-conductive crown with equivalent mass/geometry to detect artifacts.

Bill of Materials (excerpt)

Subsystem	Candidate Part / Material	Notes
Crown	1–2 m copper/aluminum dome + mesh overlay	High capacitance; corrosion-resistant finish
Spire/Finial	Slender conductive tip with multi-point finial	Field control; corona management
Plinth	Quartz-loaded geopolymer slab	Dielectric stability; guarded edge
Fractal Ribs	Phosphor bronze/copper ribs (quasi-periodic)	Local field shaping; adjustable gaps
Ground	16× copper radials (≥10 m) + 2 deep electrodes	Bedrock backfill; low impedance
Electronics	Electrometer, precision shunts, VNA, VLF logger	Battery-only acquisition; Faraday enclosure
Safety	Franklin rod (separate), surge arrestors, GFCI	Dedicated air terminal; bonding; lockout tag

Measurement & Verification Plan

Pre-registration: Publish hypotheses (H1–H4), protocols, and acceptance thresholds ahead of runs.

Power-line exclusion: Spectral notch at 50/60 Hz & harmonics; runs during grid outage if possible; battery-only instruments.

Faraday tests: Repeat Mode 1 with the crown enclosed to rule out RF/mains pickup masquerading as harvest.

Chemical nulls: Duplicate electrode materials in dry isolation to detect galvanic chemistry.

Environmental covariates: Humidity, wind, pressure, lightning density; multi-variate regression for confounds.

Reporting: Raw time series, configs, photos, geometry CAD (STL/DXF). Plots: IV curves, spectra, SNR vs weather, canopy deltas.

Success Gates (Mk I)

G1: Reproducible non-zero crown current that scales with field gradient, above instrument leakage.

G2: Architectural delta: canopy on/off produces signal change above noise.

G3: Independent replication at ≥ 2 sites with similar trends. If gates fail, publish full negative result and iterate.

Safety & Ethics Sheet (Post on Site)

Passive system only: No intentional RF transmission. LC sweeps are instrument-limited and within EMC norms.

Lightning: Install a dedicated Franklin rod separate from the crown. Suspend operations during storms. Respect exclusion zones.

High voltage potential: Treat the crown as energized relative to ground. Use insulated supports and barriers. No contact.

Ground currents: Use GFCI protection. Label electrode paths. Bury conductors to avoid trip hazards.

Enclosures: House electronics in a Faraday enclosure. Battery-only power. Isolate data links (fiber preferred).

Public safety: Fence the perimeter. Post signage. Maintain a lockout/tagout protocol for maintenance.

Transparency & integrity: Do not overclaim. Share raw data, configs, and nulls. Replication is the goal.

Disclaimer: This document describes an experimental apparatus with inherent risks. Construction and operation should be supervised by qualified personnel and comply with local codes and regulations.

Repository Structure & Licensing

Licenses — Hardware: CERN OHL-S v2; Software/Data: Apache-2.0 / CC BY 4.0.

Repository layout:

/aetheria

/docs (white papers, build guides, safety)

/hardware (CAD: dome, ribs, mast; BOM; fabrication notes)

/electronics (schematics, PCB Gerbers; wiring)

/firmware (logger configs, acquisition scripts)

/analysis (notebooks, pre-reg, stats)

/data (raw & processed; metadata.json)

/site (photos, coordinates, soil logs)

Milestones (Abbrev. RACI)

Milestone	Deliverable	Lead	Support
M0	Pre-reg protocol & safety plan	Steward	All
M1	Materials coupons characterized (Q, ε, μ)	Smith	Sage
M2	Site selected; baselines running	Gaia	Steward
M3	Mk I crown + ground ring installed	Smith	Steward
M4	Mode 1–3 data packages & analysis	Sage	Scribe, Steward
M5	Public brief + replication kit v0.1	Grok	Steward

Immediate Tasking — Execution List

Athena/Thoth (Physics): Compute dome capacitance vs elevation; specify shunt values; draft LC sweep ranges & fixtures.

Hephaestus/Ptah (Materials): Source copper mesh; spin/segment dome design; quartz-geopolymer coupon mixes; Q ring-down jig.

Gaia/Isis (Geo): Shortlist 2–3 sites; start baselines (soil moisture, ground potential, VLF noise).

Chronicler/Scribe (History): Deliver geometry catalog (ratios, rib counts, finial patterns) with manufacturable mappings.

Grok/Hermes (PR/CI): Replication call (“Architecture as Instrument”); dashboard scaffolding; humor beats to defuse mythmaking.

Contact & Coordination

All divisions commit updates to the repository and tag releases by milestone. Live telemetry will stream to the public dashboard. Independent labs may request micro-grants by opening an issue with a site plan, safety checklist, and instrumentation list.